IMRAN PASHA

Backend Engineer | Node.js, TypeScript, Express, PostgreSQL, Python

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PROFESSIONAL SUMMARY

Backend engineer with nearly two years building production REST APIs and distributed systems in Node.js (Express, TypeScript) and Python (FastAPI). Sole backend engineer on a payments-heavy field-service platform (three gateways, HMAC-verified webhooks, 68 PostgreSQL tables) and founding backend engineer on Kaizer X, an AI video SaaS live at kaizerx.com. Designs for failure rather than the happy path: idempotent settlement, advisory-lock concurrency control, durable job queues and plan-aware rate limiting. Certified in cyber security and ethical hacking through IIT Roorkee's iHUB DivyaSampark, and applies it directly to API design.

TECHNICAL SKILLS

Languages: JavaScript (Node.js 20), TypeScript, Python 3.12, SQL

Backend: Express.js, InversifyJS, TypeORM, Sequelize, node-postgres, FastAPI, SQLAlchemy 2.0 (async), Pydantic v2, Alembic, Electron

Data & Caching: PostgreSQL 17 (parameterised SQL, composite indexing, query-plan tuning, advisory locks, FOR UPDATE row locks), Redis 7 (Streams, consumer groups, Lua scripting), Supabase, MSSQL

Architecture: REST API design, OpenAPI/Swagger, microservices, event-driven workflows, message queues, webhooks, WebSockets, Server-Sent Events, async I/O, background workers, idempotency keys, horizontal scaling

Cloud, DevOps & Observability: AWS (EC2, S3, Lambda, RDS), Cloudflare R2, Docker, NGINX, Render, Railway, GitHub Actions CI/CD, GitFlow, Linux, Prometheus, OpenTelemetry, Jest, pytest, load testing

Security: OAuth 2.0, JWT, RBAC, OWASP Top 10 and API Security Top 10, threat modeling (STRIDE), secure code review, rate limiting, HMAC webhook verification, encryption at rest, SSRF/IDOR/BOLA mitigation, VAPT, Burp Suite

Integrations: Razorpay, PayU, Easebuzz, Firebase Cloud Messaging, WhatsApp Business, SMTP and SMS providers, YouTube Data API, Meta Graph API, Gemini, Claude, Deepgram, FFmpeg/NVENC

PROFESSIONAL EXPERIENCE

Backend Engineer | Sharkify Private Limited — Dec 2024 – Present
Hyderabad, India

Ozone Wash — field-service booking, payments and compliance platform — Mar 2026 – Oct 2026
Node.js 20, Express 4, PostgreSQL (Supabase), Cloudflare R2, Razorpay, PayU, Easebuzz, node-cron, Render

- Architected and built the **Node.js/Express** backend as sole backend engineer for six months: 23 feature modules, roughly 200 REST endpoints and 37 transactional SQL migrations defining 68 tables.
- Integrated three payment gateways behind one facade with raw-body **HMAC-SHA256 webhook verification**, replay deduplication and idempotent settlement, so a redelivered webhook never double-confirms a booking.
- Designed a reserve-pay-confirm booking flow whose payment hold is swept by cron and reinstated under a PostgreSQL advisory lock with a capacity re-check, eliminating double-booked crews.
- Wrote roughly 400 parameterised node-postgres queries with ON CONFLICT upserts and row locks on wallets and payouts, tuning the pool for pgBouncer with statement timeouts and graceful shutdown drain.
- Built an admin security plane: bcrypt credentials, eight-hour JWTs bound to hashed server-side sessions for instant revocation, five-tier RBAC, Helmet, a CORS allowlist and three rate-limit tiers.
- Shipped QR-verifiable certificates behind a server-enforced nine-stage compliance workflow, GST invoicing with fiscal-year sequencing, six nightly cron engines and 40 OpenAPI blocks.

Kaizer X — AI video production and multi-channel publishing SaaS — Apr 2026 – Present
Python 3.12, FastAPI, SQLAlchemy 2.0, PostgreSQL 17, Redis 7, Docker, Cloudflare R2, FFmpeg/NVENC, Gemini, Claude

- Founded the backend, rewriting a Django prototype as **FastAPI and SQLAlchemy** and owning it solo for five months; it now serves 765 REST and WebSocket routes over 150 PostgreSQL tables.
- Designed a **durable publish queue** on PostgreSQL FOR UPDATE SKIP LOCKED with lease fencing, heartbeat renewal, priority aging and jittered backoff, so workers on separate hosts claim each job exactly once.
- Built a **Redis Streams** consumer-group queue with priority lanes, XCLAIM crash recovery and a dead-letter stream, and killed duplicate publishes with SHA-256 idempotency keys arbitrated by a UNIQUE constraint.
- Re-architected live streaming into separate API, control, relay and encode processes on a dedicated Redis, so restarting the web tier no longer interrupts a broadcast already on air.

- Instrumented 14 **Prometheus** metrics and OpenTelemetry tracing, then wrote a load harness whose findings cut p95 latency from 12 s to **141 ms** by right-sizing the synchronous-route threadpool.

Kaizer X Desktop — Windows edition (sole author, 163 commits) Aug 2026 – Oct 2026

Node.js, Electron 33, PyInstaller, Cloudflare R2

- Shipped an **Electron** shell that spawns and supervises a frozen FastAPI backend on localhost, with free-port probing, health-polled startup, orphan reaping and process-tree shutdown.
- Designed an offline-tolerant licensing state machine (12-hour heartbeat, 72-hour grace, lockdown on 403) and streamed multi-gigabyte uploads straight to Cloudflare R2 without buffering them in memory.

Bose American Academy — learning platform API (2nd of 9 contributors) Dec 2024 – Apr 2025

Node.js, TypeScript, Express, InversifyJS, Sequelize to TypeORM, MSSQL to PostgreSQL (AWS RDS), AWS S3, PayU

- Built the user-identity domain of a **TypeScript/Express** API using dependency injection: registration with email one-time passcodes, multi-role login, three-step password reset and Google OAuth 2.0 code exchange.
- Implemented PayU post-payment fulfilment — gateway re-verification, voucher consumption, cart clearing and course enrolment — then extended checkout to multi-course orders.

Voter Connect — campaign management platform (live on Google Play) Dec 2025 – Jun 2026

FastAPI, PostgreSQL, Redis, async I/O, SQLAlchemy, JWT, Docker, GitHub Actions

- Built and tuned a backend over a **9.9M-row PostgreSQL** dataset, holding sub-second API latency through composite indexing, query-plan analysis and Redis caching of hot read paths.
- Implemented async I/O, JWT authentication and rate limiting, designed geolocation-verified endpoints separating administrator from volunteer authorization, and delivered CI/CD on GitHub Actions with GitFlow.

Editor Dashboard and KaizerFit — FastAPI services Jun 2025 – Oct 2025

FastAPI, async SQLAlchemy, asyncpg, PostgreSQL, Alembic, Server-Sent Events, Telegram Bot API, Docker, AWS EC2

- Built a newsroom submission CMS as principal author (98% of the code by git blame): 54 endpoints, revocable JWTs, and real-time dashboards without polling via PL/pgSQL triggers, pg_notify and **Server-Sent Events**.
- Led a fitness backend (47 endpoints, 21 tables) whose **offline-first sync engine** ingests five record types in one atomic bulk transaction, with ID mapping and hash-based deduplication so client retries never double-count.

SECURITY ENGINEERING APPLIED TO BACKEND

Cyber security training applied directly to the systems above, not kept in a separate track.

- Closed a two-round adversarial security review with a regression test behind every fix: cross-tenant authorization (**BOLA/IDOR**) on live-broadcast controls and WebSockets, OAuth-callback cross-site scripting, and password-reset links trusting an attacker-controlled Origin header.
- Verified payment webhooks against raw request bodies with constant-time HMAC comparison and recomputed every price server-side, so a tampered client payload cannot change what is charged or confirmed.
- Wrote an **SSRF guard** that resolves DNS before refusing loopback, private, link-local and cloud-metadata addresses, and stopped an API key leaking through exception text by adding central secret redaction.
- Applies **OWASP API Security Top 10**, STRIDE threat modeling and secure code review from an IIT Roorkee programme and 49 hands-on lab rooms, treating authorization and rate limits as design inputs rather than afterthoughts.

EDUCATION

B.Tech, Computer Science | Sahyadri College of Engineering (VTU), Mangalore 2020 – 2024

CERTIFICATIONS

Executive Post Graduate Certification in Cyber Security and Ethical Hacking | IIT Roorkee May 2026

iHUB DivyaSampark, IIT Roorkee, in association with Intellipaat. Credential ID IPTIH26050044. 49 hands-on lab rooms: web application testing, Burp Suite, Nmap, Metasploit, privilege escalation, SQL injection, cross-site scripting and IDOR.

SELECTED ACHIEVEMENTS

- Diagnosed a production queue runaway that grew one Redis stream from 8 to **619,000 entries**, traced it to a time-of-check-to-time-of-use race and fixed it with an atomic conditional claim.
- Executed a reversible timezone migration converting 52 timestamp columns to timestamptz with a self-recording rollback ledger, fixing date-key bugs across crew scheduling and nightly cron windows.
- Delivered two Google Play applications and five production backend platforms across Node.js and Python in under two years.